Advancements Report - Automated Annotation of Primate Vocalizations

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Repository: https://github.com/nhrot-fc/tesis-primate

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1. The Problem

The researchers study 8 primate species in the Peruvian Amazon, listed in Table 1.

Table 1. Target species. The two-letter code identifies the species throughout this report; classes are labelled `code/call_type`.

Code	Common name	Scientific name
AA	Night monkey	*Aotus azarae*
AC	Peruvian spider monkey	*Ateles chamek*
AS	Howler monkey	*Alouatta sara*
CC	Shock-headed capuchin	*Cebus cuscinus*
LW	Weddell's saddleback tamarin	*Leontocebus weddelli*
PT	Toppin's titi monkey	*Plecturocebus toppini*
SB	Bolivian squirrel monkey	*Saimiri boliviensis*
SM	Large-headed capuchin	*Sapajus apella macrocephalus*

Data comes from AudioMoths deployed across the forest: non-intrusive and continuously recording. The result is thousands of hours of audio that must be reviewed manually.

A researcher opens each file in Raven, draws a time–frequency box around every vocalization, and labels it with a species and a call type. Recording speed exceeds review speed by a wide margin, so most of the audio is late or never analyzed.

Operating requirement. The deliverable is a triage tool: the model proposes regions, an expert confirms. Missing a vocalization costs more than reviewing a false positive, so every design choice is biased toward **recall over precision**.

The team currently uses **Arbimon** for pattern-recognition-based event retrieval, with decent results but the syllables have variations and the tool only receives a single reference. Additionally, the tool does not produce time–frequency box annotations with species and call type, readable back into Raven. That is the gap this work targets.

2. State of the Art

Detecting **time–frequency** boxes is repeatedly called underexplored — Zhu & Sato (DCASE 2025) state that "the detection of time-frequency bounding boxes remains largely unexplored." The mainstream bioacoustic tools support this: BirdNET (Kahl et al., 2021) scores fixed 3 s segments and Perch 2.0 (van Merriënboer et al., 2025) is a clip-level classifier over ~14,600 species; neither localizes calls in frequency.

Three loosely connected lines of work exist:

- **2D boxes, transformer detectors (DETR family)**: Zhu & Sato (DCASE 2025 Workshop) pair a self-supervised AST backbone (EAT) with DINO ("DETR with Improved DeNoising Anchor Boxes", Zhang et al., ICLR 2023), evaluated not on bioacoustics but on whistle sounds from defective wind-turbine blades (AP50 0.494 vs. 0.365 for a Faster R-CNN/ResNet baseline). Likely the first self-supervised-AST + deformable-DETR for 2D SED, but not the first DETR-based 2D detector: Cotillard et al. (2024, JASA) fine-tuned DETR on spectrograms for beluga whale calls, and SEDT/SP-SEDT (Ye et al., 2021) applied DETR to SED in 1D.
- **2D boxes, region-CNN / YOLO detectors**: DeepSqueak (Coffey et al., 2019, *Neuropsychopharmacology*) pioneered Faster R-CNN on rodent ultrasonic-vocalization spectrograms; extended to multi-species soundscapes (SILIC, Wu et al., 2022), marine mammals (Hamard et al., 2024, Faster R-CNN + FPN) and bats (Bat Detective, 2018; BatDetect2, 2022, which predicts time, duration, frequency range and species per call). Most recently Hexeberg et al. (2026, arXiv:2606.10407) apply YOLO11 to dense tropical soundscapes (Singapore, with out-of-distribution testing on Hawaii; 81.8% vs. 42.1% IoMin@50 F1 in-distribution) and adopt **Intersection-over-Minimum (IoMin)** as a matching metric tolerant of ambiguous acoustic boundaries — presented as their contribution, but mathematically the Szymkiewicz–Simpson overlap coefficient, already used in vision as "IoM" (ReMOTS, 2020; Vogel et al., 2023). A conceptual precursor is Kong et al. (2019, IEEE/ACM TASLP): weakly-supervised time–frequency segmentation masks rather than boxes.
- **1D boxes (time only)**: YOHO (Venkatesh et al., 2022), Sound Event Bounding Boxes (Ebbers, Germain, Wichern & Le Roux, Interspeech 2024) and Voxaboxen (Mahon et al., DCASE/NeurIPS 2025) regress onsets/offsets on the time axis; despite the "box" terminology they carry no frequency extent.

These lines cite each other only sparsely — Zhu & Sato are the main bridge, connecting the region-CNN bioacoustics work to the 1D DETR lineage — leaving cross-domain, frequency-aware detection open. That is where this work sits.

3. Old Method

The code for this stage is not in the repository, so the numbers below are reported as recorded at the time and are not reproducible here.

Scope: **one species/call type (LW/CS).** 103 recordings, 682 annotations, fixed 0.5 s windows. Two binary classifiers - a small custom CNN and ResNet50V2 with ImageNet weights - both reached about **99% test accuracy**.

Why it was abandoned. The output is presence per window, not an event: no box, no frequency extent, no event count, no species or call-type distinction. The accuracy is also misleading. 91.2% of the clips are negative, so a constant "no call" predictor already scores 91.2%, and the split was over shuffled clips, so overlapping clips from the same annotation appeared in both train and test. Table 12 contrasts this method with the one proposed in section 5.

4. Dataset

19,029 curated annotations, 8 species, 65 distinct (species, call type) pairs. All Raven `.txt` files are unified into a single table by `domain/annotations.py`: call codes normalized through a synonym map (`contact call`, `contact syllable`, `cs_a`, `CS` → `cs`), species taken from the source directory, high frequency clipped to the Nyquist limit (22,050 Hz), and degenerate boxes (< 10 ms or zero bandwidth) dropped. 41 pairs belong to the vocabulary declared in `src/domain/species.py` and are enumerated in Table 2; the remaining 24 pairs (336 records, 1.8%) fall outside it and are flagged `requires_review`.

Annotated call-type vocabulary

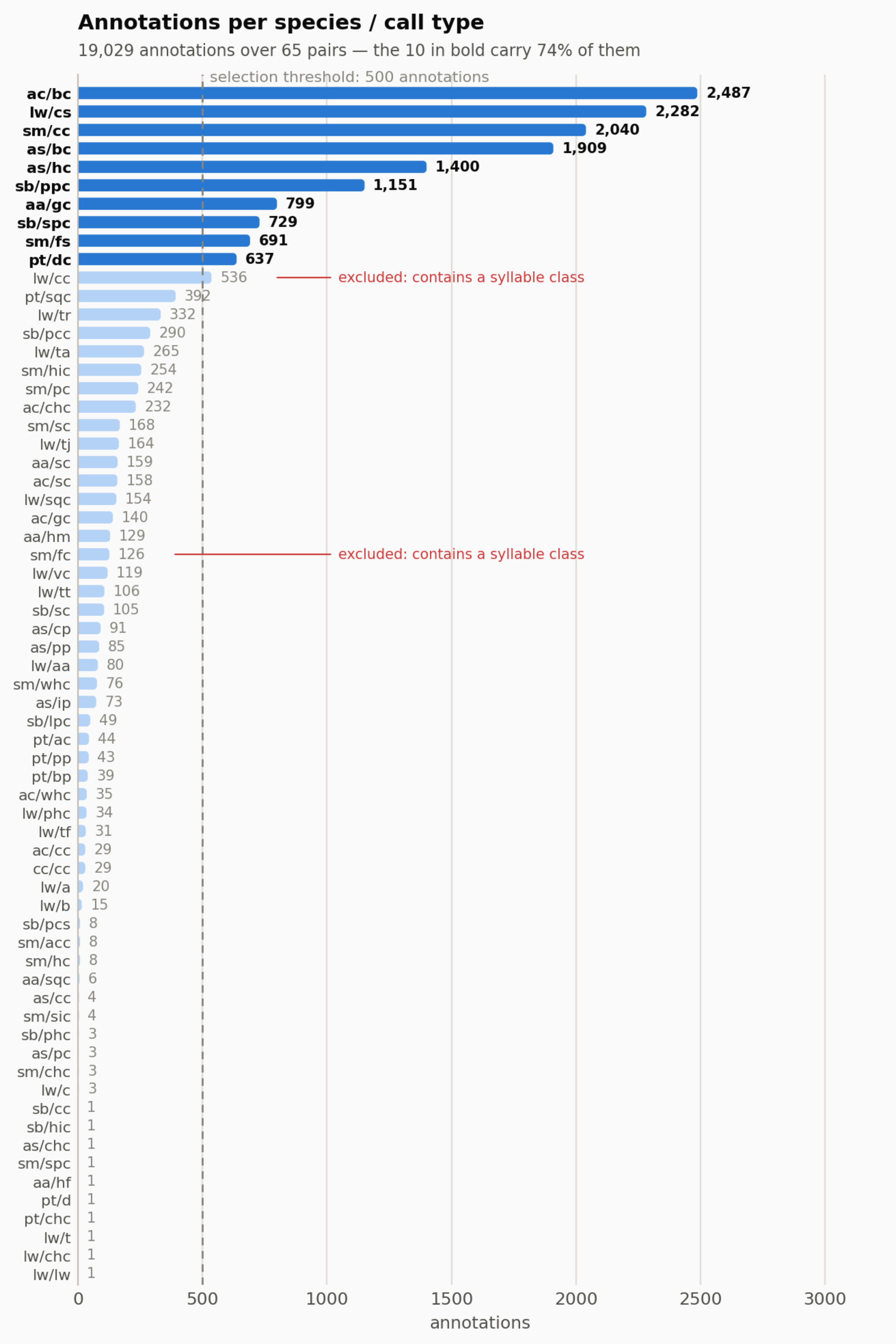
Table 2. The 41 in-vocabulary (species, call type) pairs, grouped by species and ordered by frequency. `Code` is the normalized Raven string; `Call type` the name declared in `src/domain/species.py`. `Role` marks the 10 pairs kept for the experiment subset and the two phrase classes excluded for nesting. Counts sum to 18,693; the remaining 336 of 19,029 are out of vocabulary.

Species	Code	Call type	Annotations	Role
AA	gc	gulp call	799	experiment
AA	sc	squeak call	159	tail
AA	hm	hoot call	129	tail
AC	bc	bark call	2,487	experiment
AC	chc	chitter call	232	tail
AC	sc	squeak call	158	tail
AC	gc	growl call	140	tail
AC	whc	whinnie call	35	tail
AC	cc	contact call	29	tail
AS	bc	bark call	1,909	experiment
AS	hc	howl call	1,400	experiment
CC	cc	contact call	29	tail
LW	cs	contact syllable	2,282	experiment
LW	cc	contact call	536	excluded (phrase)
LW	tr	trino call	332	tail
LW	ta	terrestrial alarm call	265	tail
LW	tj	unofficial tj call	164	tail
LW	sqc	squeal call	154	tail

Species	Code	Call type	Annotations	Role
LW	vc	visual contact call	119	tail
LW	tt	trino transition call	106	tail
LW	aa	aerial alarm call	80	tail
LW	phc	phee call	34	tail
LW	tf	trino fast call	31	tail
PT	dc	duet call	637	experiment
PT	sqc	unofficial squeal call	392	tail
PT	ac	alarm call	44	tail
PT	pp	unofficial pant phrase	43	tail
PT	bp	unofficial bellow phrase	39	tail
SB	ppc	play peep call	1,151	experiment
SB	spc	spit call	729	experiment
SB	pcc	peep contact call	290	tail
SB	sc	shriek call	105	tail
SB	lpc	long peep call	49	tail
SM	cc	contact call	2,040	experiment
SM	fs	food syllable	691	experiment
SM	hic	hip call	254	tail
SM	pc	purr call	242	tail
SM	sc	squeal call	168	tail
SM	fc	food call	126	excluded (phrase)
SM	whc	whistle call	76	tail
SM	acc	aggressive contact call	8	tail

Call types labeled with "Unnoficial" are call types not found on the reference PPT, could be typos or new call types. They are not included in the experiment subset.

Two properties of Table 2 drive the rest of this section. First, the vocabulary is uneven per species: LW carries 11 distinct call types over 4,103 annotations, while CC contributes a single call type with 29. Second, the distribution within a species is as long-tailed as the distribution across species - `sm/acc` has 8 annotations against `sm/cc`'s 2,040, a 255× spread inside one species. Figure 1 shows the same distribution pooled across all pairs.



Annotations per species / call type
19,029 annotations over 65 pairs — the 10 in bold carry 74% of them
selection threshold: 500 annotations
ac/bc 2,487
lw/cs 2,282
sm/cc 2,040
as/bc 1,909
as/hc 1,400
sb/ppc 1,151
aa/gc 799
sb/spc 729
sm/fs 691
pt/dc 637
lw/cc 536
excluded: contains a syllable class
pt/sqc 392
lw/tr 332
sb/pcc 290
lw/ta 265
sm/hic 254
sm/pc 242
ac/chc 232
sm/sc 168
lw/tj 164
aa/sc 159
ac/sc 158
lw/sqc 154
ac/gc 140
aa/hm 129
sm/fc 126
excluded: contains a syllable class
lw/vc 119
lw/tt 106
sb/sc 105
as/cp 91
as/pp 85
lw/aa 80
sm/whc 76
as/ip 73
sb/lpc 49
pt/ac 44
pt/pp 43
pt/bp 39
ac/whc 35
lw/phc 34
lw/tf 31
ac/cc 29
cc/cc 29
lw/a 20
lw/b 15
sb/pcs 8
sm/acc 8
sm/hc 8
aa/sqc 6
as/cc 4
sm/sic 4
sb/phc 3
as/pc 3
sm/chc 3
lw/c 3
sb/cc 1
sb/hic 1
as/chc 1
sm/spc 1
aa/hf 1
pt/d 1
pt/chc 1
lw/t 1
lw/chc 1
lw/lw 1
0
500
1000
1500
2000
2500
3000
annotations

Figure 1. Annotation count per (species, call type) pair. A handful of pairs carry most of the data

Size variation on Calls

The calls in this dataset are very uneven in size, and that fact drives the design decisions that follow.

Table 3. How short or long, and how narrow or wide, the annotated calls are, measured over all 19,029 annotations.

	shortest	longest	ratio
Duration	0.034 s	73.3 s	**2,134×**
Bandwidth	224 Hz	21,784 Hz	97×

The longest call lasts more than 2,000 times longer than the shortest one. For comparison, in COCO (the standard image collection used to benchmark object detectors) the biggest objects are only about 10 to 100 times the size of the smallest ones.

This has two consequences. No single clip length works for every class: a window long enough to hold a 73 s howl is far too coarse for a 34 ms chirp.

Syllables and phrases

A *phrase* is a sequence of *syllables*, and the experts box both: 94.2% of `lw/cc` phrases contain at least one `lw/cs` syllable, and 96.8% of `sm/fc` phrases contain an `sm/fs` syllable. So the same audio carries one big box and several small boxes inside it.

The model cannot represent that. It returns a flat list of boxes, and training pairs each true box with exactly one predicted box, with no way to say "this box contains that one". Keeping both levels would mean asking the model to report the same sound twice, as two unrelated events sitting on top of each other. The two phrase classes are therefore dropped before training (`EXCLUDED_PAIRS` in `create_dataset.py`); they appear in Table 2 marked *excluded (phrase)*.

Geometric separability

Species separate largely by frequency band: AS sits inside ~25–1,410 Hz (1st–99th percentile), while `lw/cs` is never annotated below 3,687 Hz. **Call types within a species** do not separate: in LW, the `t` family plus `vc` and `cs` share band and duration almost exactly. Figure 2 makes both effects visible at once. This is why detection is trained class-agnostic and evaluated separately for classification.

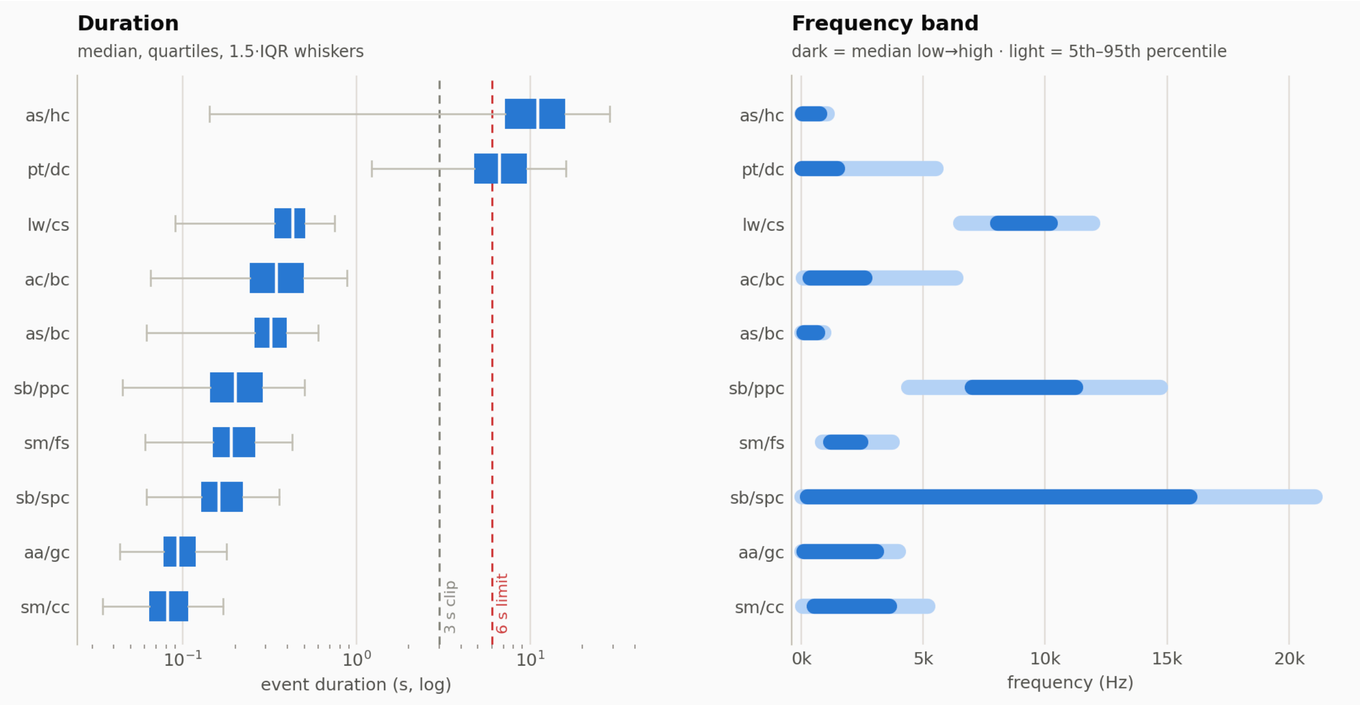


Figure 2. Where each class lives in time and frequency. Species occupy distinct bands; call types of the same species overlap almost completely - the dissociation that splits the evaluation protocol of Table 8 into detection and classification.

Two labels are not one shape: `ac/bc` (bandwidth CV 0.60) and `sm/fs` (0.63) each split into two clearly separated clouds in duration × bandwidth, and `sb/ppc` (0.47) is a single diffuse smear, as shown in Figure 3. These are the classes flagged for clustering in section 8.

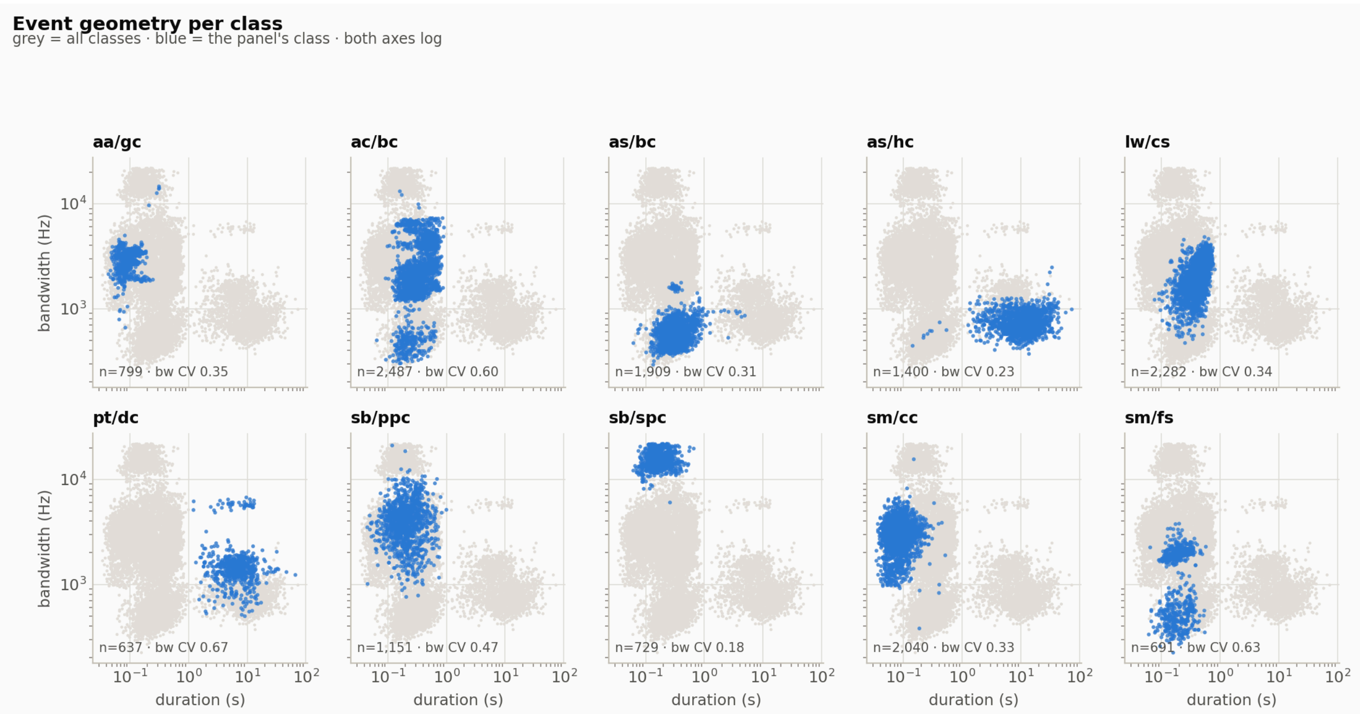


Figure 3. Duration versus bandwidth, one panel per experiment class. The split clouds of `ac/bc` and `sm/fs` suggest a single label covering two call variants; both are among the worst-recall classes in Table 11.

Experiment subset

Pairs with ≥ 500 annotations, after removing the two nesting phrases → **10 classes, 14,125 annotations** (the rows marked *experiment* in Table 2).

Splitting is **by recording file** (`split_manifest`, seed 42), so overlapping windows from one recording never cross splits - ratios **70 / 10 / 20**. Table 4 gives the resulting composition.

Table 4. Split composition after windowing. `recordings` counts source `.wav` files, `windows` the 3 s clips retained, `boxes` the annotation instances assigned to a window.

split	recordings	windows	boxes
train	1,166	8,022	16,674
val	166	1,043	2,183
test	335	2,490	5,343

There are more boxes than annotations because a single annotation can land in two windows: the 3 s window advances by 1.5 s. Training boxes per class range from 3,273 (`lw/cs`) down to 320 (`as/hc`); Figure 4 breaks the totals down by class.

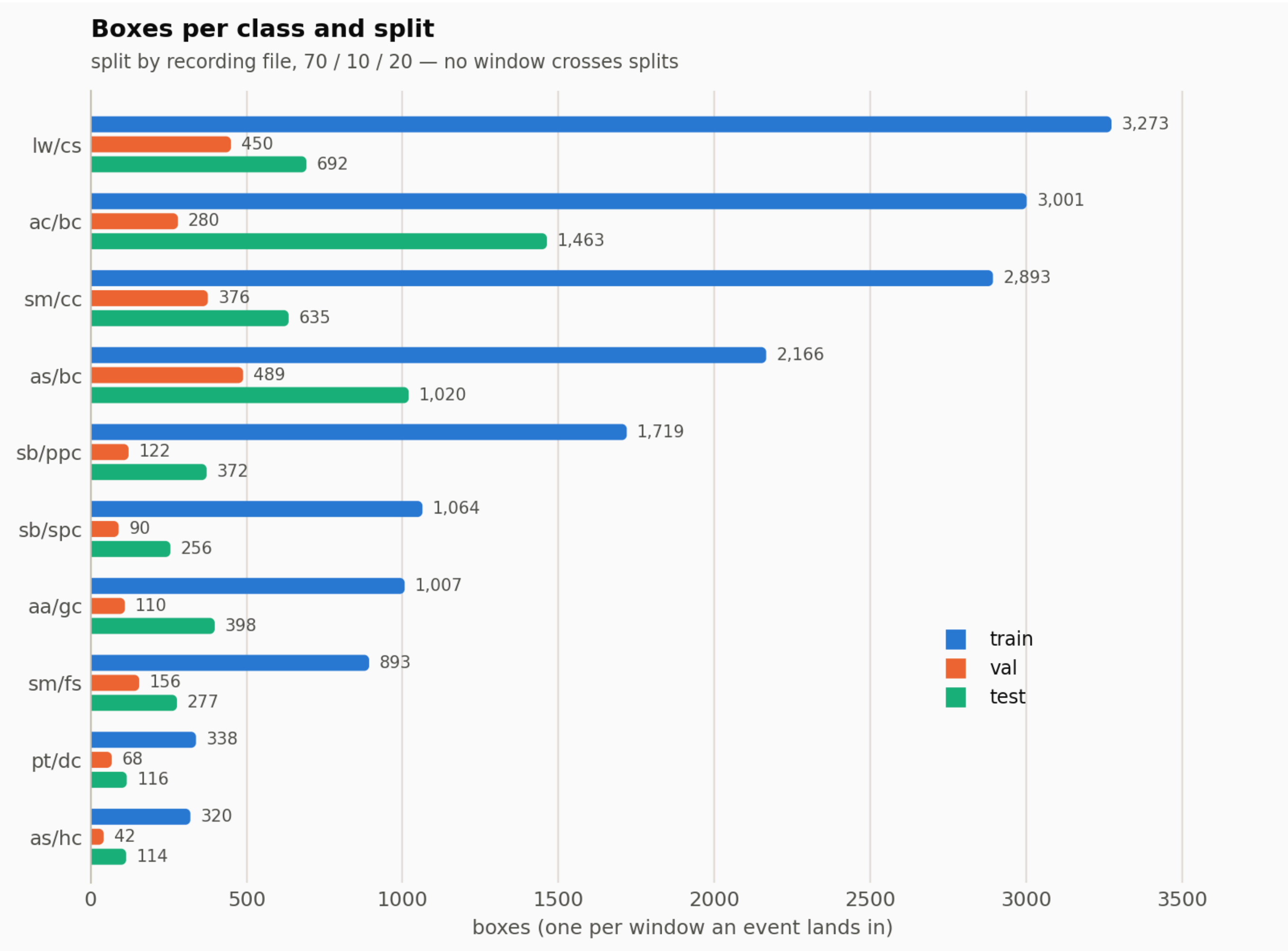


Figure 4. Boxes per class in each split. Balance is preserved across splits, but `as/hc` and `pt/dc` contribute far fewer boxes than their Table 2 counts suggest - the windowing defect quantified in Table 5.

Long events are dropped

`as/hc` and `pt/dc` produce far fewer boxes than they have annotations, and the reason is structural. A window keeps an event only if at least `min_overlap = 0.5` of the event's duration falls inside it, so **no event longer than 2 × 3 s = 6 s can ever be assigned to any window**. That removes **1,514 annotations (10.7% of the experiment subset)** before training begins, concentrated in the two classes of Table 5.

Table 5. Annotations made unreachable by the 3 s windowing. An event is unreachable when it exceeds 6 s, making `min_overlap = 0.5` unsatisfiable at every window position. No other experiment class loses a material share.

class	annotations	unreachable (> 6 s)	share lost
as/hc	1,400	1,147	82%
pt/dc	637	367	58%

A further 13.5% of the subset is longer than the 3 s window and survives only as a box clipped to the window edge. Figure 5 places these losses against the classes that are unaffected. There is no separate long-bout code path today; this is the main open item in the preprocessing stage.

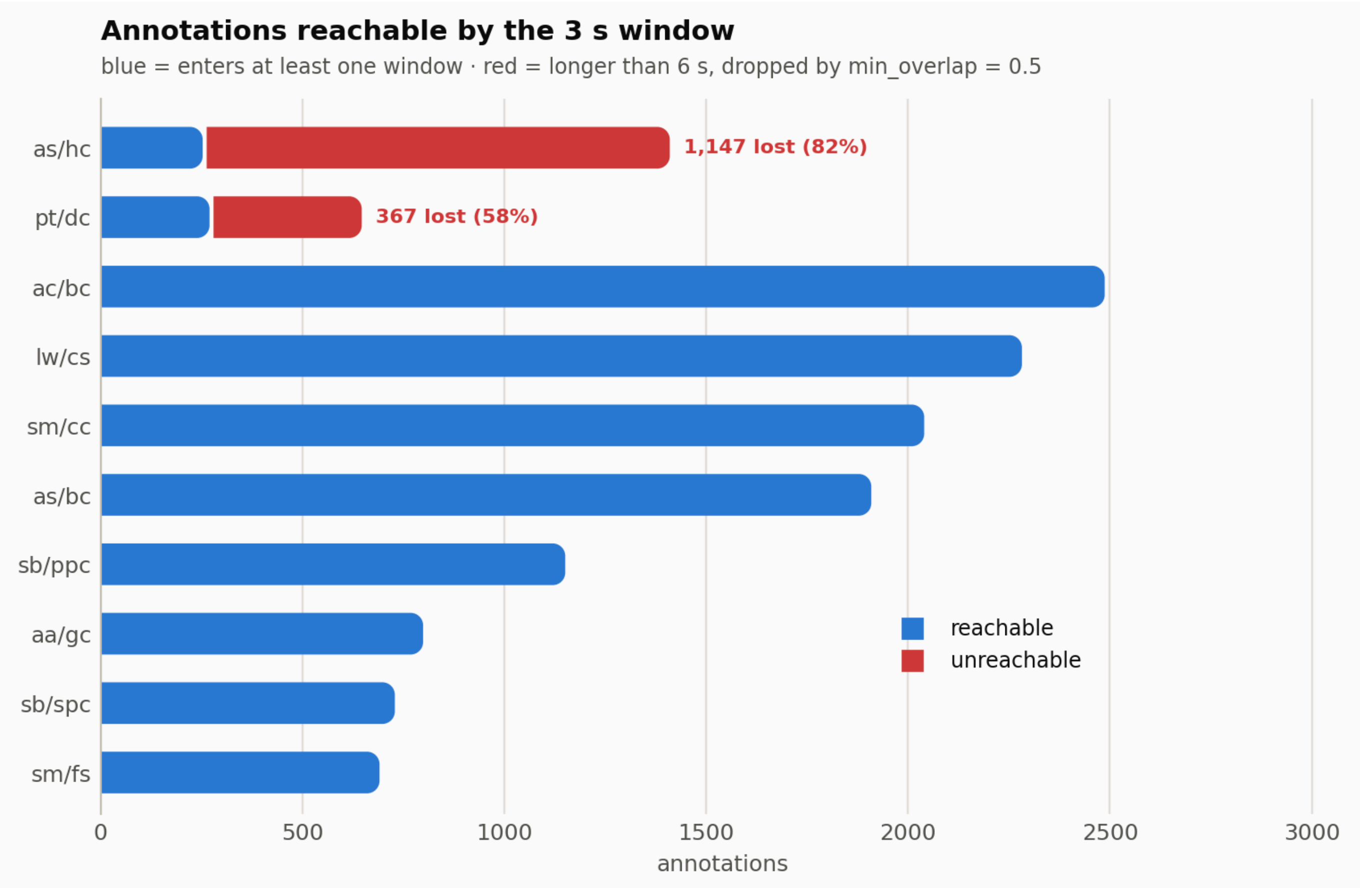


Figure 5. Share of annotations each class can contribute under the 3 s windowing. For `as/hc` most of the label is invisible to training, capping its recall in Table 11 regardless of model capacity.

5. New Method

Audio Spectrogram Transformer (AST) backbone + Deformable DETR decoder. In four steps:

1. The audio is turned into a log-mel spectrogram - an image with time on one axis and frequency on the other, exactly what the analyst sees in Raven.

2. A **frozen AST** pretrained on AudioSet converts that image into features, and a ViTDet-style pyramid rebuilds them at four resolutions, so both a 34 ms chirp and a 3 s bout are visible at some scale - the range documented in Table 3.
3. A **Deformable DETR decoder** carries 64 learnable queries. Each query attends to a few sampled points of the pyramid and emits one candidate event.
4. During training, **Hungarian matching** pairs every true annotation with exactly one query. The model therefore learns to output a *set* of events directly - no anchors, no NMS inside a clip, no post-hoc peak picking.

Each prediction is a tuple `(t_start, t_end, f_low, f_high, species, call_type, score)` - exactly what an analyst draws in Raven, so predictions are written back as a selection table.

88.9 M parameters total, of which 3.1 M are trainable (dim 128, 64 queries, 10 classes); the 85.8 M AST parameters are frozen.

Preprocessing

Table 6. Preprocessing parameters, as set in `Parameters` (`src/core/config.py`). The rationale column records what failed at the alternative setting.

Parameter	Value	Rationale
Clip / hop	3.0 s / 1.5 s	Fits 8 of the 10 classes; see the long-event defect in Table 5
`min_overlap`	0.5	An event enters a window only if half its duration is inside
`win_length`	1024 (23 ms)	Shorter than the shortest events (34 ms min, 81 ms median for `sm/cc`)
`n_fft`	4096	Zero-padding only: densifies the frequency grid without touching temporal resolution
`hop_length`	400 (9 ms)	Gives 331 frames per clip
`n_mels`	128	What AST expects; 256 gave no measurable gain
`f_min` / `f_max`	25 Hz / 22,050 Hz	`f_max` at 16 kHz was truncating half of `sb/spc`'s boxes (mean high freq 16,110 Hz)

Boxes are normalized to `cxcywh` in `[0,1]`, with the frequency axis mapped through the HTK mel scale (`hz_to_y`), so a box means the same thing in pixel space as on screen. Figure 7 shows the result for one window.

Architecture decisions

Table 7. Deviations from the base Deformable DETR recipe, each with the failure it addresses. Their cumulative effect is measured in Table 10.

Decision	Justification
AST frozen	Using pretrained weights avoids the need to train from scratch

Decision	Justification
`time_stride` 10 → 5	The shortest events are 34 ms. Each token is now 45 ms instead of 91 ms. The model can now better localize shorter events.
Zhu et al. attention init	Sampling offsets start on a radial grid instead of noise
Iterative box refinement + auxiliary losses	Six decoder layers were sharing one fixed reference point
True multi-scale sampling	Feature map 12×64 → pyramid 48×256, 24×128, 12×64, 6×32; each level is sampled separately
`n_queries` = 64	At most 11 boxes land in one clip (median 2, Figure 6); 100 queries produced prediction noise
`eos_coef` 0.1 → 0.05	Recall bias: lowers the cost of predicting an event where there is none
Decoder dropout 0.1	No regularization existed

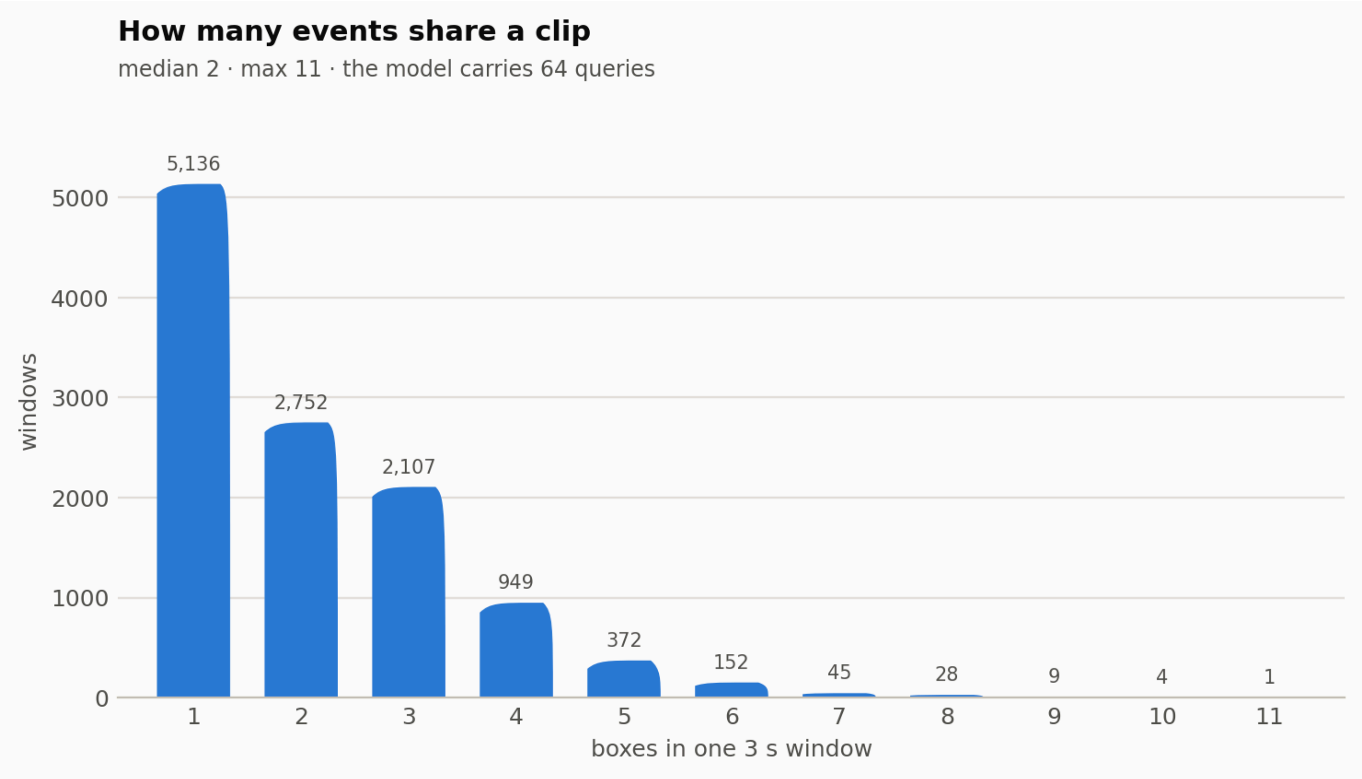


Figure 6. Annotation boxes per 3 s window: median 2, maximum 11. This is the basis for the query budget of 64 in Table 7 - well above the worst case, without inviting spurious predictions.

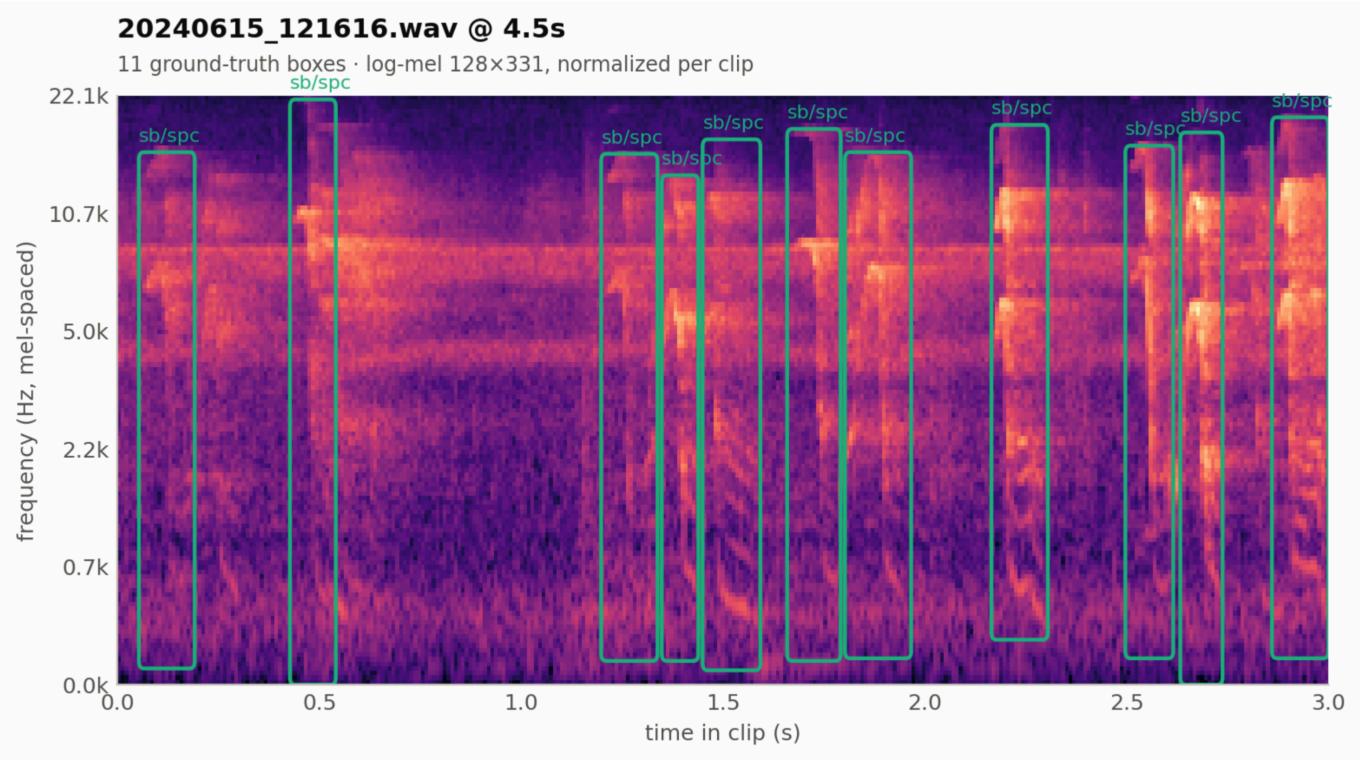


Figure 7. One training window as the model receives it: a 3 s log-mel spectrogram (parameters of Table 6) with the target boxes overlaid in the normalized `cxcywh` frame.

Evaluation protocol

Three failure modes are measured separately, because they have different fixes.

Table 8. Evaluation protocol. The three tasks are scored independently so a drop points at a specific stage: detection is a recall problem, framing a regression problem, classification a feature problem. Table 9 follows this decomposition.

Task	Question it answers	Metric
Detection	Did we find the event at all?	class-agnostic recall @ IoU 0.5, score threshold 0.5
Framing	Is the box drawn tightly around it?	mean IoU over matched pairs, plus AP @ {0.25, 0.30, 0.50, 0.75}
Classification	Is the label right?	accuracy over matched pairs

Plus **FP/TP at the operating point**, which is the human review burden - the number the domain expert has to approve. Checkpoint selection maximizes an F-β with β = 3, i.e. it weights recall nine times more than precision, per the operating requirement in section 1.

All reported metrics use plain IoU. IoMin (Hexeberg et al.) is implemented in `architectures/iou.py` and selectable as the matcher/loss criterion, but is not yet reported alongside IoU; doing so would quantify annotation-boundary ambiguity. GIoU and EIoU were implemented and removed - EIoU is unbounded below zero and not comparable to published work.

Per-file inference (`pipelines/inference_pipeline.py`) does apply class-wise NMS at IoU 0.3, but only to merge duplicate detections of the same event across the overlapping sliding windows - not to resolve predictions within a clip.

6. Results

Table 9. Validation-set performance of `128d_64q_iouiou_10cls`, along the three axes of Table 8. `FP/TP` is measured at the operating point (score threshold 0.5) and is the review burden the expert experiences directly.

Axis (per Table 8)	Metric	Value
Detection	class-agnostic recall @ IoU 0.50	0.767
Detection	FP/TP at the operating point	1.12
Framing	mean IoU over matched pairs	0.659
Framing	class-agnostic AP @ 0.25	0.803
Framing	class-agnostic AP @ 0.50	0.648
Framing	class-agnostic AP @ 0.75	0.133
Classification	accuracy over matched pairs	0.982

In plain terms: out of every 10 vocalizations, the model proposes about **8**; the expert discards roughly **one wrong proposal per correct one**; and when a proposal is correct, its species and call type are right **98%** of the time. What it does not yet do well is draw the box *tightly* - that is the AP@0.75 = 0.133 figure.

On real recordings

Figures 8–14 use the same audio files for both models, so they can be read side by side: Figure 9 and Figure 11 are the same recording, and Figure 10 and Figure 12 are the same recording.

Old model - a probability curve per window (Figures 8–10):

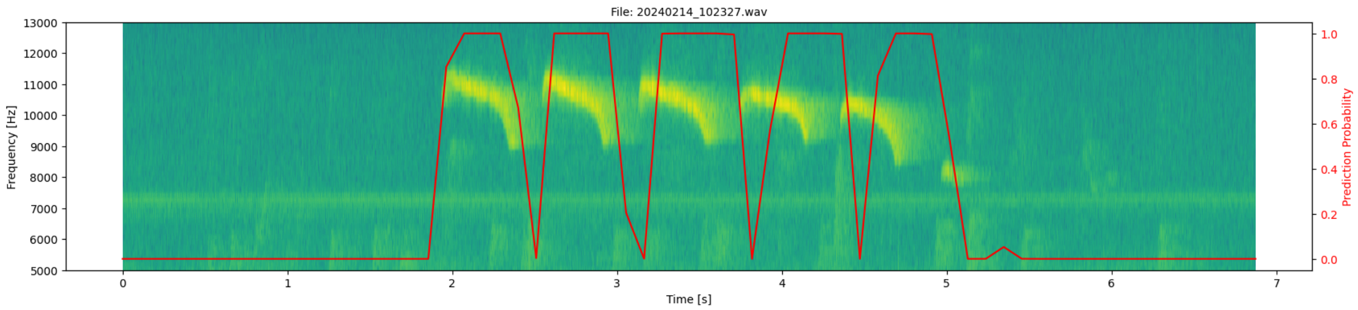


Figure 8. Old classifier on `20240214_102327.wav`. The red curve is per-window call probability. It rises on the six LW/CS syllables, but the output is a curve, not events: no box, no frequency extent, no class, and a syllable count only where the curve dips back to zero.

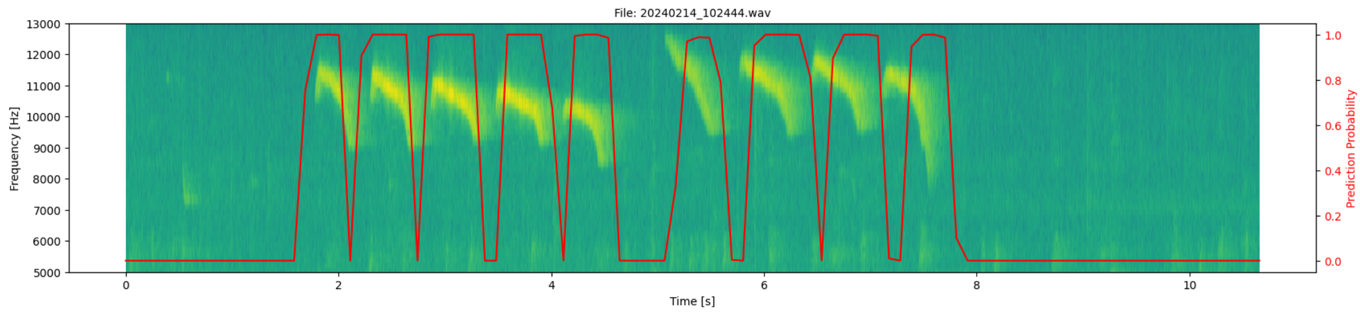


Figure 9. Old classifier on `20240214_102444.wav`. Ten syllables, curve saturated at 1.0 through most of the bout; where it fails to return to zero, adjacent syllables merge into one detected region. Compare Figure 11, same recording under the new model.

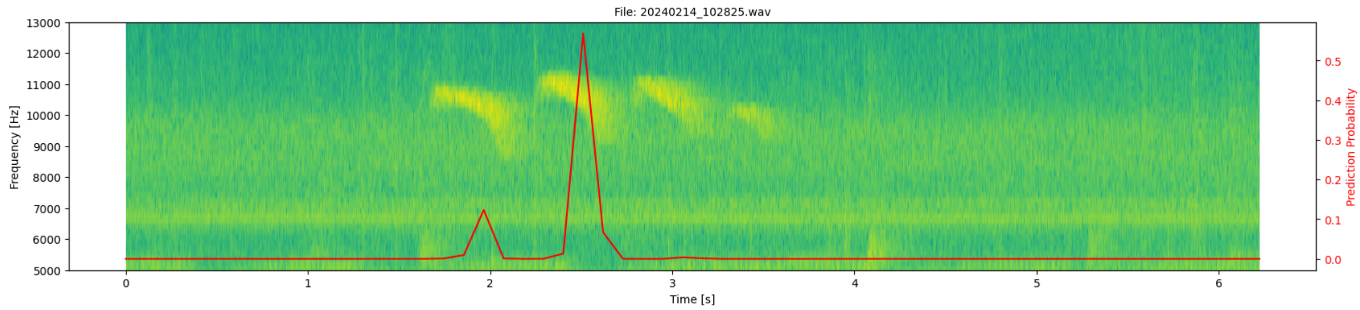


Figure 10. Old classifier on `20240214_102825.wav`, a quieter recording of the same call type. Probability barely exceeds 0.55 and peaks once, between two syllables: four visible LW/CS calls are effectively missed. Figure 12 shows the same file under the new model.

New model - a set of labelled boxes (Figures 11–14):

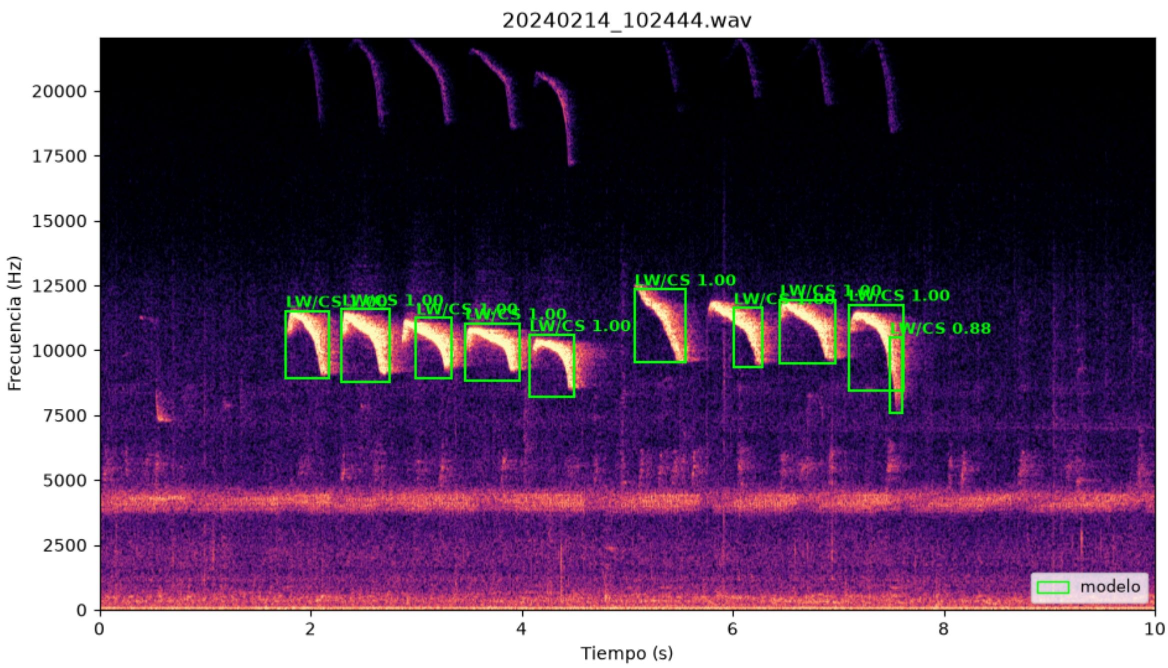


Figure 11. New model on `20240214_102444.wav`, the recording of Figure 9. Each of the ten LW/CS syllables comes back as its own box with class and confidence (mostly 1.00). The harmonic arcs above 17 kHz are correctly left unboxed.

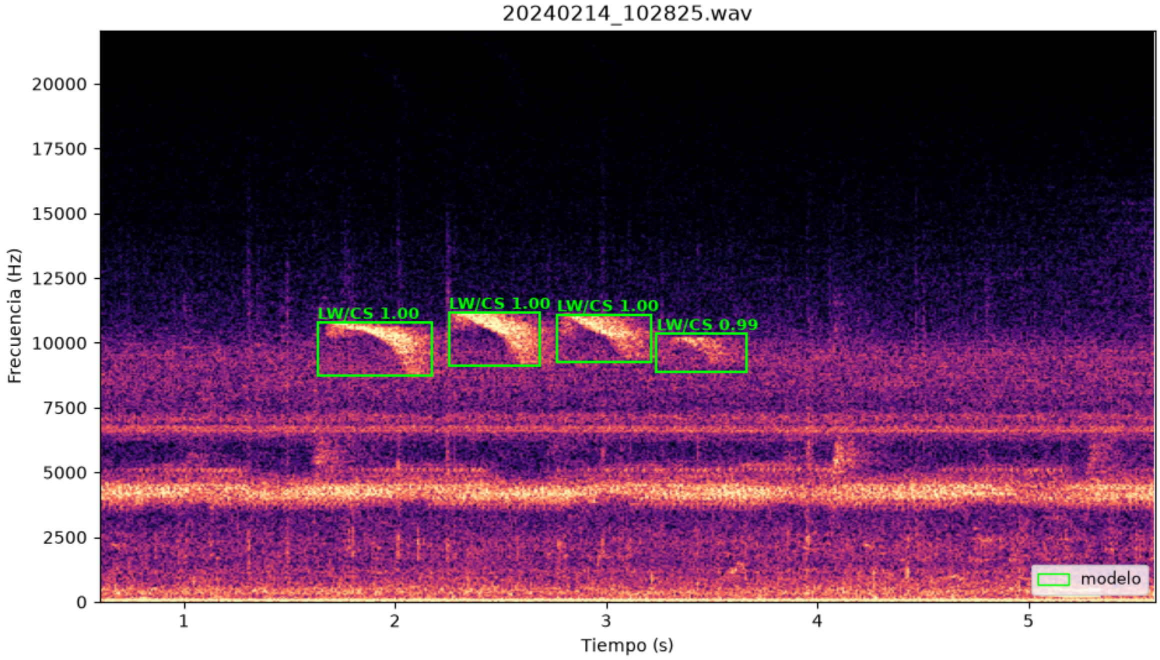


Figure 12. New model on `20240214_102825.wav`, the recording the old classifier missed in Figure 10. All four LW/CS syllables are found with confidence ≥ 0.99, despite the strong broadband noise floor below 7 kHz.

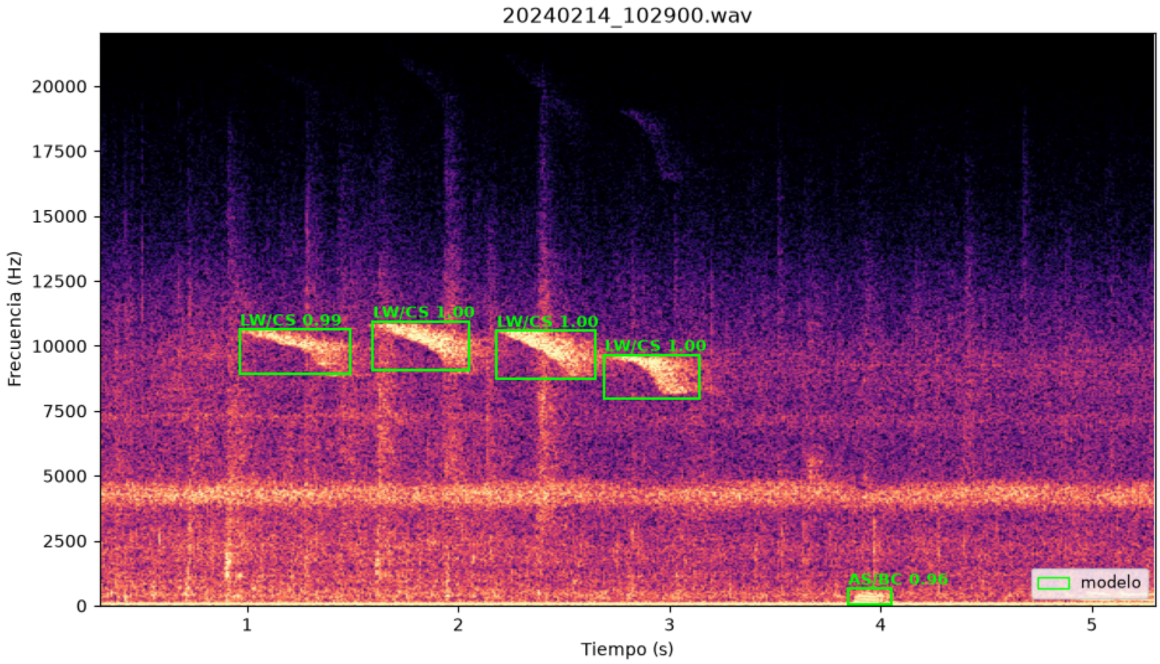


Figure 13. New model on `20240214_102900.wav`, with heavy rain/insect streaks. Four LW/CS syllables near 10 kHz plus one AS/BC event below 1 kHz at ~3.9 s - two species in different bands retrieved in the same pass, the separability of Figure 2 at inference time.

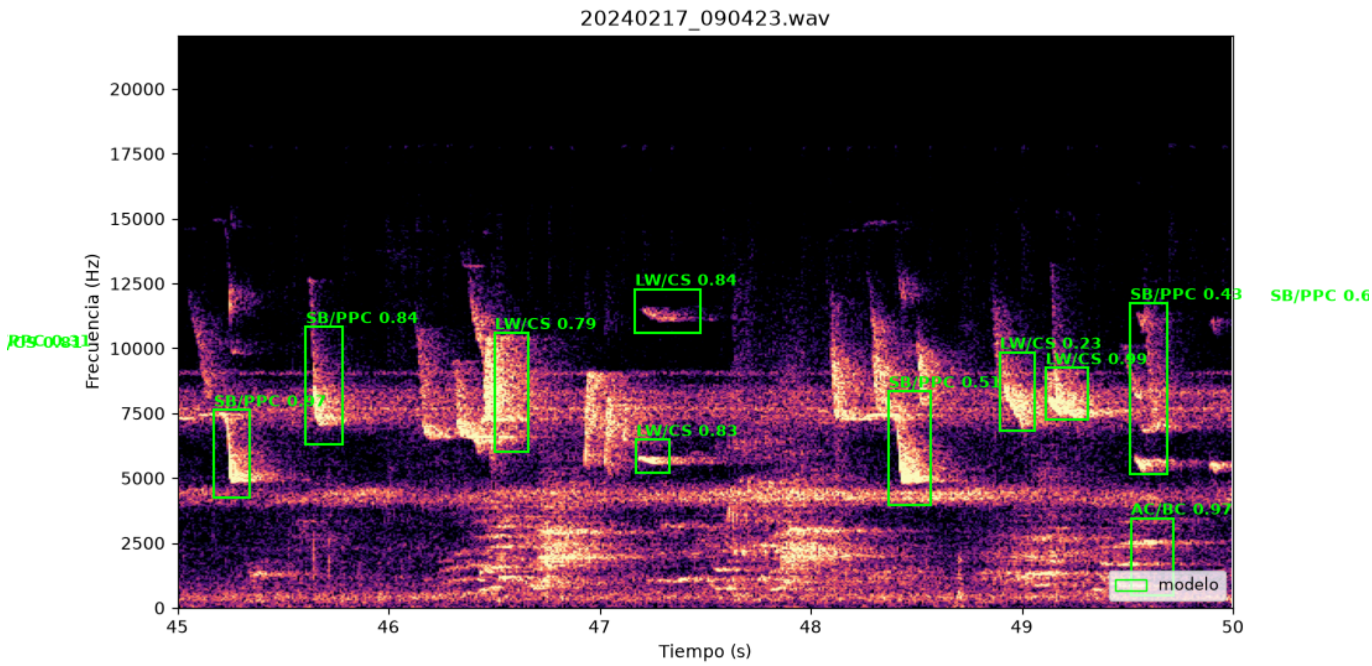


Figure 14. New model on `20240217_090423.wav`, seconds 45–50: three species (SB/PPC, LW/CS, AC/BC) overlapping, some events cut at the segment edge. Confidences spread (0.23–0.97) and box edges are loose - in this dense regime framing, not detection, is the bottleneck, consistent with the AP@0.75 of Table 9.

Progression

Table 10. Effect of the architecture changes of Table 7 on class-agnostic AP, each row cumulative over the one above. `epochs` is where the reported checkpoint was selected. Bold marks the configuration of Table 9.

Configuration	AP@0.25	AP@0.50	AP@0.75	epochs
Base Deformable DETR	0.609	0.325	0.037	30
+ attention init, iterative refinement, aux losses	0.742	0.569	0.112	75
+ IoU in matcher and loss	**0.803**	**0.648**	**0.133**	**20**

The last change improved every threshold while cutting training to a quarter of the epochs.

Per-class recall

Table 11. Class-agnostic detection recall @ IoU 0.5 by ground-truth class, for the configuration of Table 9. `note` attributes each shortfall to a cause established earlier: unreachable long events (Table 5), geometric heterogeneity (Figure 3), or nesting (section 4).

class	recall	note
aa/gc	0.891	
as/bc	0.883	
lw/cs	0.851	

class	recall	note
as/hc	0.762	only 253 of 1,400 annotations are reachable; the rest exceed 6 s (Table 5)
sm/cc	0.721	81 ms median duration
sb/ppc	0.689	geometrically heterogeneous (bandwidth CV 0.47, Figure 3)
pt/dc	0.649	only 270 of 637 annotations are reachable (Table 5)
sm/fs	0.622	nested inside `sm/fc`, which is excluded (Table 2)
sb/spc	0.556	its band spans 249–15,931 Hz and swallows several other classes
ac/bc	**0.473**	**heterogeneous (bandwidth CV 0.60, Figure 3), worst despite 3,001 training boxes**

7. Comparison with the proof of concept

Table 12. The proof-of-concept classifier of section 3 against the detection model of section 5, on the properties that determine whether the output is usable as a Raven annotation. The two are not comparable on a single metric, hence two quantities in the metric row.

	Old (POC classifier)	**New (AST + Deformable DETR)**
Task	Binary classification per 0.5 s window	Set detection over a 3 s spectrogram
Labels	LW only, call type merged into one positive class	7 species × call type (10 classes)
Output	Presence probability	Boxes (time *and* frequency) + class + confidence
Overlapping calls	Not separable	Separable; event count is native
Metric	Clip accuracy (91.2% baseline)	Event-level recall / FP-TP at IoU, per class
Raven export	Not accurately implemented	Implemented

8. Next Steps

In progress

1. **Long-event handling.** 10.7% of the experiment subset is unreachable under the current windowing (Table 5). Either a second, coarser clip length for `as/hc` and `pt/dc`, or a relaxed `min_overlap` with edge-clipped boxes.
2. **Data augmentation.** Random temporal offset in windowing (targets position memorization), gain jitter ±10 dB, box jitter ±15% relative to size, and background mixing from unannotated segments of the team's own recordings.
3. **Report IoMin alongside IoU** in `training_pipeline.evaluate`.

Requires the research team

4. **Expert review of high-confidence false positives.** 48.9% of them fall on regions with no annotation at all. Whether these are noise or unannotated events changes the interpretation of every precision figure in Table 9.
5. **Inter-annotator agreement.** Two experts annotating the same recordings gives the achievable performance ceiling, which is what separates "the model fails" from "the task has a limit". Currently unmeasured.
6. **Cluster `ac/bc` and `sb/ppc` geometry.** If two or three clean modes appear (Figure 3), the label groups distinct call variants - a finding about the annotation scheme rather than the model.
7. **Extend the label set beyond the 10 experiment classes** of Table 2. 29 in-vocabulary call types are currently unused, and 24 out-of-vocabulary pairs (336 records) are pending curation.
8. Benchmark against Arbimon on the same recordings.